Bureaucrats in Congress: The Politics of Interbranch Information Sharing

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Congress often relies on bureaucrats' information for policy production. However, scholars lack an empirical understanding of what drives information sharing between bureaucrats and legislators. We argue that the partisan alignment between a bureaucrat and legislator determines the amount and type of information transmitted. Using new comprehensive data on bureaucratic witnesses in committee hearings, as well as a new measure of the informational content of testimonies, we show that less analytical information is transmitted between a bureaucrat and legislator pair when the legislator is a presidential out-partisan rather than a co-partisan and that this effect is heightened when the bureaucrat is a political appointee. At the aggregate hearing level, the collective amount of analytical information from bureaucrats is lower under divided government than unified government but is offset by the analytical information from nonbureaucratic witnesses. These dynamics provide a nuanced understanding of the information transmission between bureaucrats and Congress.

The relationship between Congress and the executive agencies is central to the separation of powers in American politics: Congress makes laws, and the executive branch executes the laws through its system of bureaucratic agencies. Scholars have studied how partisanship across this separation of powers system affects outcomes, such as legislative productivity (e.g., Binder 1999; Coleman 1999; Howell et al. 2000; Phillips and Kirkland 2018), oversight and investigations (Kriner and Schickler 2016; Kriner and Schwartz 2008), and the degree of delegation to the bureaucracy (Epstein and O'Halloran 1996, 1999).

Beyond these outcomes, however, lies a broader information asymmetry between the two branches: The division of responsibilities leads each side of the interbranch relationship to gain different information and perspectives about policymaking (Banks and Weingast 1992; Bendor, Taylor, and Gaalen 1987). Bureaucrats in executive agencies—who implement policy—acquire a relatively higher awareness and on-the-ground knowledge about the realities of policy implementation and its consequences compared to lawmakers in Congress. While Congress also can collect this information from others during the legislative process, bureaucrats remain one of their best sources of expertise about policy implementation and execution (Gailmard 2002; Gailmard and Patty 2012).

How, then, do partisan divides across the legislative and executive branches influence interbranch information sharing? We argue that the partisan alignment between legislators and bureaucrats drives the amount and type of interbranch information exchange. When a legislator and bureaucrat pair is misaligned in committee hearings—that is, when a legislator is of one party and the bureaucrat is from an executive

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branch controlled by the other party—the flow of policy-relevant information may be impeded by the strategic interactions between legislators and bureaucrats who are both conscious of their partisan goals and political incentives.

Since the agenda and all other specific aspects of a committee hearing are determined by committee members, and witnesses respond to questions from legislators, the overall policy discourse that occurs in hearings is predominantly shaped by legislators. Legislators may ask different questions that elicit different types of information based on whether they are interacting with bureaucratic witnesses whose preferences are aligned with their partisan goals. For example, a legislator may take the opportunity to score partisan points through grandstanding and increased political messaging or aggressiveness (Park 2021, 2023). This incentive may be stronger when a legislator is questioning a bureaucrat from an executive branch controlled by the opposite party. The amount and type of information exchanged between a bureaucrat and legislator may thus be influenced by how the legislator, driven by the partisan alignment between herself and the bureaucrat, elicits this information.

Simultaneously, bureaucrats have their own incentives that steer their interactions with Congress and may play a similar role in determining the amount and type of information shared with Congress. Bureaucrats' behavior in sharing policy expertise with politicians may also depend on who they are specifically interacting with from a bureaucrat's perspective. Bureaucrats may provide more policy-relevant information to assist presidential co-partisans than they do to presidential out-partisans, or they may provide more valuable information to out-partisans to convince out-partisan legislators. Furthermore, the partisan alignment between bureaucrats and legislators may be more salient for politically appointed bureaucrats compared to careerists given the partisan nature of appointments. Thus, the amount and type of information provided to legislators also may be influenced by how the bureaucrat, driven by partisan alignment between herself and the legislator, responds to information requests.

Combining the incentives of legislators and bureaucrats, it is not clear ex ante how partisan alignment would affect information transmission between a bureaucrat and legislator at the pair level. We investigate this empirically and examine information sharing between bureaucrats and Congress in one major venue through which Congress can request, receive, and publicly disseminate information from the bureaucracy: committee hearings. Previous work has focused on hearings specific to oversight or investigative matters (Kriner and Schickler 2016; McGrath 2013). Beyond the oversight context, hearings can facilitate the exchange of information for legislative matters more broadly and, specifically, reveal the content of the information provided to Congress (Quirk 2005). While there are other channels through which legislators and bureaucrats can transmit information, such as in phone calls, written correspondence, or private meetings, hearings represent a public, formal channel of information transfer. This venue is an impactful setting for information transmission and dissemination, as the information is observed by other legislators, bureaucrats, and officials in both branches of government as well as monitored by stakeholders in the policy area.

Not all information is equal. Using House committee hearing transcripts from 1997 to 2014, we apply a crowdsourced supervised learning method to capture a specific aspect of information conveyed in witness testimonies. While informational content can be characterized in various ways, we focus on a type central to technical policy development and applicable across issue areas: falsifiable statements about the policy under consideration, which we term analytical information. This approach improves on prior measures by leveraging human judgment to assess information intensity in context rather than relying on binary classifiers (Esterling 2011) or researcher-selected word lists that may be incomplete (Ban, Park, and You 2023). Specifically, we (1) employed large-scale human coding of sample testimonies using online workers, (2) trained multiple machine learning algorithms to predict analytical information levels, and (3) optimized their combined predictions to measure analytical content in 981,633 statements from 14,092 hearings. We then conducted extensive statistical and substantive validations of this new measurement.

We use this new measure of information in combination with a novel dataset on bureaucratic witnesses from 1977 to 2014. We collect the federal agency affiliation and appointment type for each bureaucrat who testified in all House committee hearings during this extensive time span as well as the content of their testimonies and their responses to questions from members of Congress in all committee hearings. The result is a dataset that, for the first time, provides the name of the agency of each bureaucrat who appeared in a congressional hearing (1977–2014) and links to the content of information bureaucrats provided to Congress for a subset of the time period (1997–2014). Furthermore, while previous work on bureaucrats in hearings has been limited to aggregated counts of hearings or witnesses at the Congress level, we create pair-level data at the politician–bureaucrat level by parsing hearing transcripts, which allows us to analyze dyadic interactions between a member and a witness.

At the pair level between a legislator and bureaucrat, we find significantly less analytical information is transmitted from bureaucrats to legislators when bureaucrats face an

out-partisan legislator compared to a legislator of the same party of the president. This lower transmission of analytical information between misaligned legislator–bureaucrat pairs occurs in both legislative hearings and oversight hearings and after controlling for the legislator's questioning style, which we capture by the amount of grandstanding, analytical keywords used, and sentiment displayed. Using bureaucrat fixed effects, we also find that the same bureaucrat provides significantly less information to legislators who are presidential out-partisans compared to presidential co-partisans in the same hearing. The effect of partisan alignment is more salient in hearings on more polarized issues, among political appointees—especially those from executive departments—in oversight hearings, and in more prestigious and policy-oriented committees. The effect is further amplified for the agencies that are more aligned with the president. A placebo test with nonbureaucratic witnesses shows that these information-sharing dynamics are not present for other types of witnesses, suggesting that partisan alignment has a distinct effect on bureaucrats, which leads them to change their information provision to Congress.

In addition to the pair-level exchange of information between a bureaucrat and legislator, we also examine the effects of partisan alignment at the overall hearing level. Even if information exchange is impeded when a bureaucrat faces a legislator from a party opposite to the president's, in any given hearing, there are other witnesses who may also provide analytical information. Does partisan alignment between the legislative and executive branches drive the collective amount of analytical information shared between the branches at the hearing level? We use the presence of divided government as a measure of partisan alignment between the legislative and executive branches and capture the total number of question and answer pairs between legislators and witnesses as well as the total amount of analytical information provided in a hearing. We find that during divided government, members of Congress ask more questions of witnesses, a behavior that is primarily driven by presidential out-partisan members and especially when they question nonbureaucratic witnesses. Our results show that while the total amount of analytical information from bureaucrats is lower under divided government compared to unified government, this negative effect of divided government is offset by the analytical information provided by nonbureaucratic witnesses.

Altogether, this article reveals the complex nature of the relationship between partisan alignment across government and interbranch information sharing. Our findings provide a nuanced understanding with implications not only for how partisan alignment affects the information transmitted between pairs of legislators and bureaucrats, but also how it affects legislators' behavior with nonbureaucratic witnesses and the collective amount of analytical information gained by Congress.

Expertise and information are critical sources of the bureaucracy's power. While there has been ample theoretical attention devoted to information asymmetries between the two branches (e.g., Banks and Weingast 1992), how bureaucrats' selection and learning can affect the development of expertise within the executive branch (e.g., Gailmard and Patty 2012), and how bureaucrats may act strategically in response to Congress (e.g., Lowande 2019; Potter 2019), there has been relatively little attention paid to the resulting provision of information from the bureaucracy to Congress. Our findings contribute to the literature by revealing how the pair-level partisan incentives of legislators and bureaucrats and the presence of divided government affect how much policy-relevant information is shared from the bureaucracy to Congress in both legislative and oversight contexts. As information is the input to policy, our findings shed new light on how the interbranch relationship can affect the policymaking process vis-à-vis information flows.

INTERBRANCH INFORMATION SHARING

The information advantage that bureaucrats have concerning program implementation is a crucial factor in the canonical power balance between Congress and the bureaucracy. While traditional delegation models, such as Huber and Shipan (2002), focus on how Congress can influence this power balance by choosing the amount of delegation Congress gives to the bureaucracy, another way to influence the interbranch relationship is by controlling information. It has been theorized that the institution that has more information about the costs and consequences of policy implementation holds an "informational advantage" in this interbranch relationship (Banks and Weingast 1992; Bendor et al. 1987).

What do we mean by policy-relevant information or expertise? Bureaucratic expertise has been measured in various ways. For instance, Clinton et al. (2012) measures the policy expertise of federal bureaucrats in each agency using the proportion of technical and proportion of professional employees. As another example, Richardson, Clinton, and Lewis (2018) use a survey approach and ask federal bureaucrats to rate their workforces from which they construct a measure of skill and competency for each agency. Their work shows that expertise varies across agencies and implies that it varies across bureaucrats within agencies as well. In this article, we focus on a type of information that is closely linked to policy expertise and central to technical policy development: information that uses falsifiable statements on the policy under consideration, which we term *analytical information* (more

explanation is provided in the "Hearing Transcript Data and a New Measure of Analytical Testimony" section

Scholars have documented that bureaucrats, who are responsible for the implementation and evaluation of their agencies' programs and policies, have deep familiarity and expertise specific to their agency's jurisdiction. This work has focused on how bureaucrats both bring expertise to and develop expertise on their jobs. Research has argued that politically appointed bureaucrats bring high levels of human capital, responsiveness, and energy to the executive agencies (Moe 1985). Career bureaucrats, especially those who have advanced through the ranks, are seen to possess subject area expertise and public management skills (Helco 1977), and research shows that this translates into higher federal program performance (Gallo and Lewis 2012; Lewis 2007). Furthermore, Gailmard and Patty (2012) emphasize the ways bureaucrats learn and acquire expertise on the job, saying that "bureaucrats are not born with all the skills they need" and that their expertise is gained through incentives in public service.

Regardless of where bureaucratic expertise originates, it remains constant in the literature that bureaucrats possess expertise and an informational advantage over Congress. Recent research has shown that politicians do, in fact, seek to obtain and rely on this information and that politicians' preferences can indeed be shaped by how bureaucrats frame an issue (Blom-Hansen, Baekgaard, and Serritzlew 2020) and the ideological alignment with bureaucrats (Bellodi 2023; Esterling 2009).

Given that bureaucrats hold the informational advantage and Congress can benefit from this information when producing policy, how is information shared between the bureaucracy and Congress? Congress can request information from bureaucrats through a variety of methods, including phoning or writing to agencies (Lowande 2018; Ritchie 2018, 2023; Ritchie and You 2019), but one formal, public way Congress requests and receives information from the bureaucracy is through committee hearings (Park 2017; Quirk and Bendix 2011). In hearings, Congress can examine the amount of information bureaucrats hold and request specific pieces of that information. Committees call bureaucrats to testify at hearings and answer questions; they use the power of subpoena, if necessary (Heitshusen 2017). Perhaps the most conspicuous way bureaucrats appear in hearings is when committees conduct oversight of the bureaucracy. Unsurprisingly, scholars have used oversight hearings as a measure of the amount of oversight that committees conduct (Kriner and Schickler 2016; McGrath 2013).

Outside of studies that examine the frequency of congressional oversight on agencies, there is limited work on the information exchange between bureaucrats and members of Congress. May, Koski, and Stramp (2016) find that bureaucrats' testimony is an important conduit of expertise and that issue maturity and salience affect the supply and demand for this expertise. Their analysis, however, is limited to hearings specifically concerned with critical infrastructure protection. Ban et al. (2023) find that when committees hold legislative hearings, they invite fewer bureaucrats during periods of divided government and substitute for them with witnesses from think tanks and universities. Eldes, Fong, and Lowande (2024), in analyzing the content of oversight hearings, show that oversight hearings can be informational and confrontational at the same time and that confrontation decreases when the legislator shares the same partisanship with the president. Bellodi (2023) analyzes legislators' citation of bureaucratic information in floor and committee speeches and shows that ideological differences and agency independence are important factors determining how often members of Congress cite the information provided by bureaucrats.

However, the existing research neglects a fundamental question about the supply of information presented to the legislature from the executive branch: What determines the information flow from the bureaucracy to Congress? The answer to this question requires examining the incentives of two groups of actors: legislators and bureaucrats. The behaviors legislators employ when requesting information from bureaucrats can determine the amount and type of information they elicit. Similarly, the way a bureaucrat provides information in response may affect the amount and type of information transmitted to the legislators. We go beyond what the previous literature has documented: We provide empirical evidence for how partisan alignment between the legislature and the executive branch influences legislator and bureaucratic behavior in committee hearings, thus shaping interbranch information flows.

PARTISAN ALIGNMENT AND INTERBRANCH INFORMATION FLOWS

We consider how partisan alignment at the individual level may shape the transmission of information between bureaucrats and legislators in committee hearings. Due to heightened partisan competition for majority status in Congress as well as the White House, legislators look for any opportunity to tarnish the image of the other party in order to make the next election favorable to their own party (Lee 2016). A legislator questioning a bureaucrat from an executive branch controlled by the other party is presented with an opportunity to target the other party through messaging. Thus, it is likely that the presidential out-partisan legislator may ask questions in a more aggressive tone or increase the political messaging in their questioning to criticize the other party or to enhance the

legislator's own partisan reputation. While the legislator still could request information to assist in policymaking, the incentive to target the bureaucrat presenting the interest of the other party for political points may dominate. When facing a bureaucrat from an executive branch controlled by their shared party, however, the legislator has less of an incentive to aggressively attack the bureaucrat and more of an incentive to request information that can assist in forming a policy of their shared preference.

The literature on the bureaucracy also suggests that bureaucrats can also act strategically. Bureaucratic subversion, shirking, and sabotage are well documented (Brehm and Gates 1997; Gailmard 2002), and bureaucrats have been shown to strategically use procedural tools, such as the timing of final rule publication, to avoid oversight by Congress and other political actors (Potter 2017, 2019). We adapt this view of strategic bureaucrats to congressional hearings and their interactions with legislators. Bureaucrats may be motivated by partisan incentives to be more willing to assist Congress—providing expertise that allows legislators to develop effective policy—if the bureaucrats feel that Congress would use their information to pass a shared policy platform. If this is the case, then we would observe more analytical information exchanged between a partisan-aligned bureaucrat and legislator pair.

When a bureaucrat is misaligned with a legislator—when the bureaucrat represents an executive branch controlled by a party opposite to the legislator's party—how much analytical information a bureaucrat responds with is less clear. On one hand, bureaucrats may not want to assist Congress in passing a diverging policy platform. When holding a legislator's question constant, bureaucrats may provide less analytical information when questioned by an out-partisan legislator than a co-partisan legislator to the president to avoid giving information that could support the opposing party's policy. On the other hand, a bureaucrat may alternatively prefer to provide more analytical information to an opposing party's legislator in an effort to still influence what the misaligned legislator decides. The effect of partisan alignment on bureaucrats is thus less clear.

The effect of partisan alignment on bureaucrats is likely more pronounced for political appointees. Bureaucrats differ significantly in their appointment status: political appointees versus career bureaucrats. Political appointees closely follow the president's party (Spenkuch, Teso, and Xu 2023), making legislators more likely to view them as extensions of the president. As a result, when out-partisan legislators seek to publicly criticize the administration, they may prioritize messaging over information-seeking during hearings with appointees compared to careerists. Additionally, political appointees operate in a different political context, as their tenure often depends on the president's continued power and favor. Their shorter-term outlook makes them more sensitive to the political environment and more responsive to the president's partisan goals (Dahlstrom, Fazekas, and Lewis 2021; Lewis 2008). Ensuring Congress advances (or blocks) legislation aligned (or misaligned) with the president may be a more pressing concern for appointees. Thus, in bureaucrat–legislator interactions, partisan alignment is likely more salient when the bureaucrat is a political appointee than a careerist.

DATA AND DESCRIPTIVE STATISTICS

In this article, we rely on two datasets. First, we construct the data on bureaucratic witnesses who appeared in hearings in the House of Representatives for the period of 1977 to 2014. We use this data to present the descriptive statistics on bureaucrats who testified in Congress. Second, we use House hearing transcripts for the period of 1997 to 2014 to analyze the transmission of analytical information from witnesses to committee members in these hearings. Using the hearing transcript data, we introduce a new measure capturing the level of analytical information conveyed in witnesses' testimonies through text-analytic methods and construct various measures for committee members' speaking tones. We then merge the two datasets and construct member–witness pair-level data for the main statistical analysis, which covers the period of 1997 to 2014.

Data on bureaucrats testifying in congress

We construct a new dataset on bureaucratic witnesses who testified in congressional hearings. We begin by using data from Ban et al. (2023), which were collected from the ProQuest Congressional, to identify witnesses who are federal bureaucrats, and we focus on hearings in the House. We then cleaned these affiliations to match them with the official name of the parent agency. We referred to various sources, such as the Office of Personnel Management (OPM) website,[1] to compile a complete list of federal government agencies and their parent organizations.[2] We performed both automated and extensive manual cleaning processes to identify the parent agency for each bureaucratic witness's affiliation.

Next, we use the OPM data to identify whether a bureaucratic witness is a political appointee or a career bureaucrat. We use the data from BuzzFeed News's Freedom of Information Act request for federal government personnel records

1. OPM website: https://www.opm.gov/about-us/open-government/Data/Apps/Agencies/.

2. We additionally checked the US Government manuals to see if the subagencies were properly assigned to parent executive departments or other agencies: https://www.govinfo.gov/app/collection/GOVMAN.

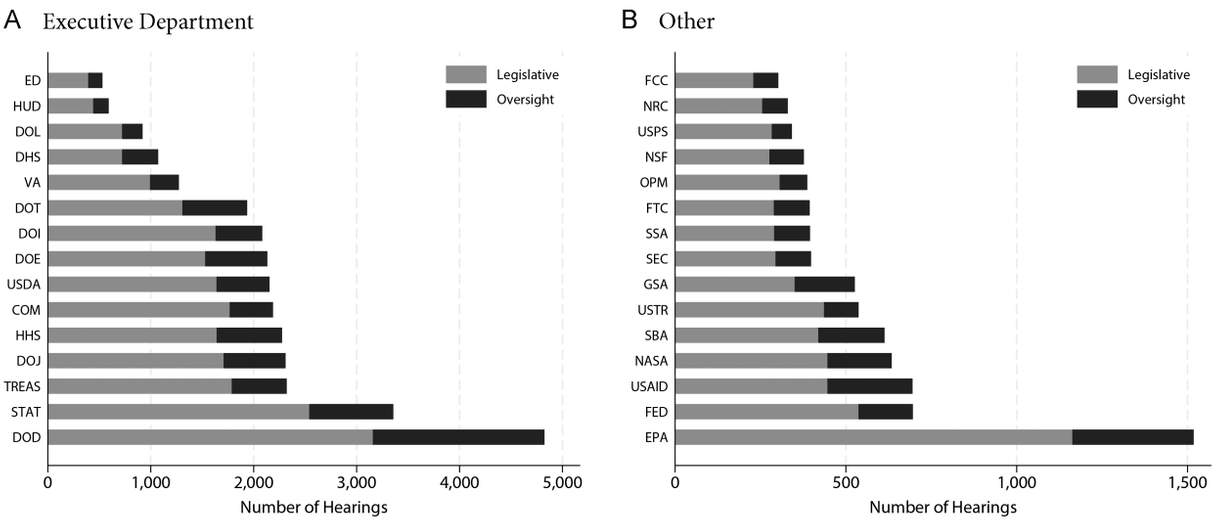


Figure 1. Legislative and oversight hearings with bureaucratic witnesses, 1977–2014.

from 1977 to 2014.[3] Following Lewis (2011), we define an individual as a political appointee if the appointment type corresponds to one of the following types in the OPM data: PAS (presidential appointments with senate confirmation), PA (presidential appointment without senate confirmation), SES (senior executive service), and C (Schedule C appointments).[4] We merged this political appointee data with the witness data by a bureaucrat's last name, first name, agency name, and year. Due to the years covered in the OPM data, our data on testifying bureaucrats spans from 1977 to 2014.

Through this process, we constructed a comprehensive dataset that includes 65,347 bureaucrat witnesses, from 15 executive departments and 55 other agencies, who testified in House hearings from 1977 to 2014. It is important to note that outside of MacDonald and McGrath (2016), who recorded the agency information for bureaucrats who testified in oversight hearings from 1999 to 2011, existing research has only tracked the frequency of bureaucrats' appearances in these hearings without examining the variations of their agency affiliations. Our dataset thus provides the first opportunity to analyze the features of bureaucratic agencies with which testifying bureaucrats are affiliated in all types of House hearings from 1977 to 2014.

In addition, our dataset includes the following hearing-level information: the name of the committee that held a hearing, the type of the hearing (legislative vs. oversight, including investigative,[5] full-committee vs. subcommittee, and referral vs. nonreferral, depending on whether a hearing refers to a specific bill or not), the type of major issue discussed in the hearing, and the total number of witnesses.

Here we provide descriptive statistics of bureaucrats who testified in hearings. Figure 1 shows the number of legislative and oversight hearings in which at least one bureaucrat testified by (*A*) the executive departments and (*B*) other agencies. These graphs show significant variation across departments and agencies. The Department of Defense has the highest frequency of hearings followed by the Departments of State, Treasury, and Justice. Among other agencies (i.e., non–executive department), bureaucrats from the Environmental Protection Agency testified most often in hearings, followed by the Federal Reserve, and US Agency for International Development. Figure 2 shows the number of career bureaucrats and political appointees testifying by executive department in panel *A* and by the top 15 other agencies who had the highest numbers of testifying bureaucrats in panel *B*. The ratio of testifying bureaucrats who are political appointees ranges

3. See https://www.buzzfeed.com/tag/opm. One limitation of this data is that it omits some agencies or individuals, such as agencies directly related to national security. Given the substantial number of bureaucrats from the Department of Defense in our dataset, we manually identified their political appointee status based on their titles that appeared in the United States Government Policy and Supporting Positions (Plum Book).

4. The OPM webpage (https://dw.opm.gov/datastandards/referenceData/1585/current?index=T) provides 18 types of appointments for federal bureaucrats. Among them, the codes 36 and 46 are PAS; the codes 55, 60, and 65 are PA; the code 44 is Schedule C appointments; and the code 50 is noncareer SES. We consider the remaining types as career bureaucrats.

5. We follow the classification by McGrath (2013), which relies on a set of keywords indicating investigative and oversight hearings and the description of a hearing in the Policy Agenda Project database. We classify hearings that are neither oversight nor investigative as legislative hearings. Among the hearings that have at least one bureaucrat appearing, 73% are legislative and 27% are oversight.

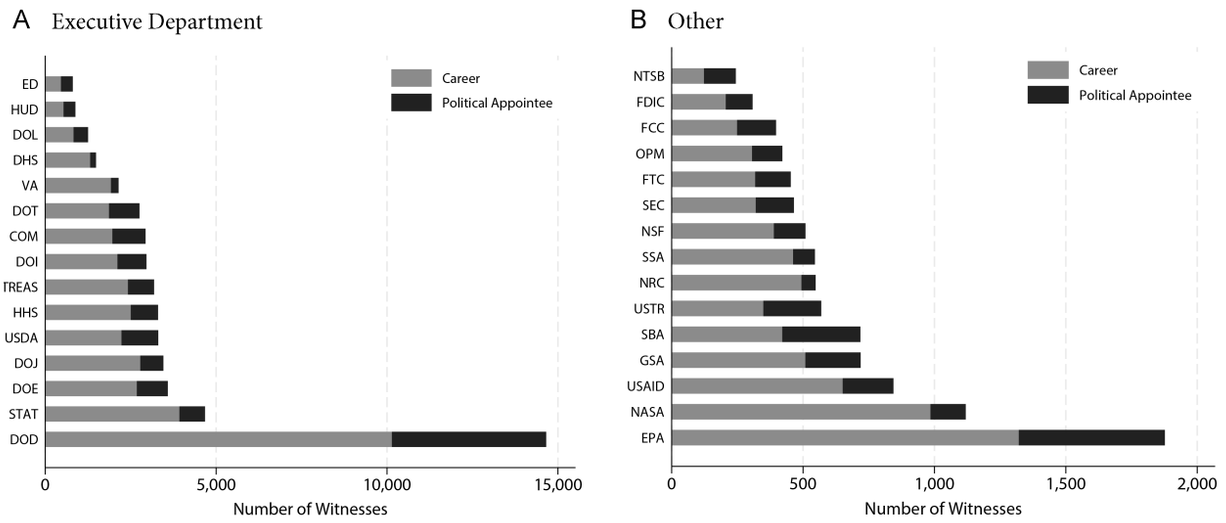


Figure 2. Career versus political appointees among bureaucratic witnesses, 1977–2014.

from 10% in Veterans Affairs to 49% in the National Transportation Safety Board.[6]

Hearing transcript data and a new measure of analytical testimony

We develop a new measure that captures the level of analytical information conveyed in witness testimonies to test our hypotheses about what impacts bureaucrats' information sharing with Congress. This measure includes all testimonies made by both bureaucrats and nonbureaucrat witnesses.

Witnesses can provide various types of information, such as an analytical, scientific analysis of the current state of a program or its potential causes and consequences, personal experiences of practitioners or those affected by a policy, or political information identifying groups benefiting from or harmed by a policy. In this study, we focus on the analytical aspect of witness testimonies for several reasons. First, previous studies have shown that legislators engage in searching for "falsifiable" or "technical" information, which we alternatively call "analytical," when making laws (Esterling 2004; Krehbiel 1991). Second, analytical information is a necessary component of technical parts of a bill. Third, as recent studies find that the analytical capacity of Congress has declined over time (LaPira, Drutman, and Kosar 2020), it is important to construct a valid measurement for analytical testimony provided by external witnesses to Congress.

To construct this measurement, we employ US House committee hearing transcripts from Park (2021) based on raw transcripts available on the Government Publishing Office website and a crowd-sourced supervised learning method that follows previously established practices (Carlson and Montgomery 2017; Park 2021).[7] This approach improves upon previous methods such as in Ban et al. (2023), which only used a dictionary of researcher-selected words. Instead, by using large-scale human coding of sample testimonies and training machine learning algorithms, we arrive at a significantly improved measure of analytical information for testimonies in Congress.

First, we define a testimony as analytical if it is (1) fact-based, (2) verifiable through research or data-driven analysis, or (3) objective. This set of concepts is largely consistent with the definition of "falsifiable" information presented in Esterling (2004).[8] Second, 3,929 sample statements were coded by online

6. In app. fig. A1 and A2, we present the over-time patterns for the types of bureaucratic witnesses and the share of political appointees.

7. A supervised learning method is preferred over unsupervised learning models when a researcher has a preset idea about how to classify texts. It provides a more delicate measure than a dictionary-based approach for two reasons. The dictionary method is (1) often agnostic about the importance of each word in relation to the concept and treats each with an equal weight and (2) has a potential problem with a homonym being used for a different meaning. Our approach resolves these issues because it relies on human judgment, which captures the concept of interest more holistically and considers the context in which a word is used through a construction of n-grams or word-embeddings.

8. To help clarify the concept, we also define what is nonanalytical. A statement is nonanalytical if it is (1) opinion-based or normative, (2) anecdotal or experiential, (3) subjective or preferential (i.e., reveals preferences of certain groups), (4) procedural statements, or (5) anything else not containing the analytical information as defined above. Our measure of analytical information is fundamentally based on the presence of falsifiable information, following other work such as Esterling (2004). However, we note that nonfalsifiable statements can still provide information that has to do with analysis and is useful in writing laws. Thus, while our measurement for analytical information is based on falsifiable information, we caution that it does not capture the full extent of information that may be analytical.

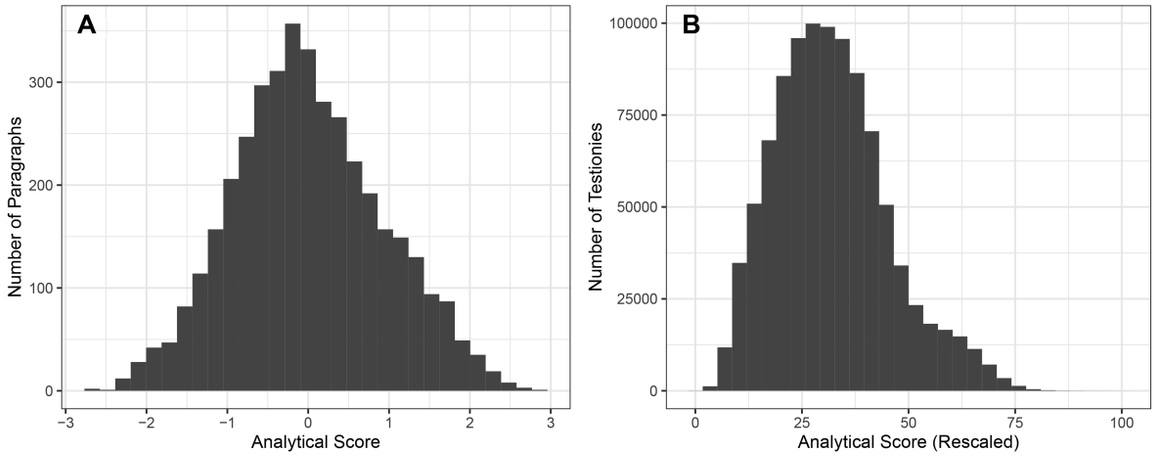


Figure 3. Distribution of the analytical score. (*A*) The distribution of the analytical score for the sample statements. (*B*) The distribution of the (rescaled) analytical score for the entire corpus of the witness testimonies for the 105th–115th Congresses.

workers at Amazon Mechanical Turk (see app. sec. D for more details about the coding process). We presented a randomly selected pair of two statements to the online workers and asked them to choose the one that was more analytical. Using their binary responses to 43,000 of these pairwise comparisons, we fit a Bradley-Terry model, which is a Bayesian model that facilitates estimating the probability that a document j will be chosen when compared with another document i by a worker k using Hamiltonian Markov Chain Monte Carlo sampling. The following equation presents the model specification and the priors for key parameters:

$$Pr(y_{ijk} = j) = \frac{exp(b_k(a_j - a_i))}{1 + exp(b_k(a_j - a_i))} \quad (1)$$

$$a_j \sim N(0,1) \quad b_k \sim trN(0,\sigma^2) \qquad \sigma \sim trN(0,3), \quad (2)$$

where N denotes the normal distribution, and trN denotes the normal distribution truncated at zero to allow positive values only. The model estimates the analytical feature of each document, a_j, and each worker's quality, b_k. The model produced a score for each sample statement ranging from -2.7 to 2.8, as shown in figure 3*A*.

Third, we randomly selected 3,500 sample statements to be used as a training set, reserved 426 statements as a validation set, and ran eight machine learning models.[9] Fourth, we constructed the final model using the ensemble Bayesian model averaging technique that assigns weights to each model to achieve the optimal prediction performance (Montgomery, Hollenbach, and Ward 2012). Using the final model, we predicted the score for the entire corpus of witness testimonies and rescaled the measurement to range between 0 and 100. In the remainder of the article, we will refer to the predicted score as the "analytical score." The graph in figure 3*B* shows the distribution of rescaled, predicted scores for all witness testimonies in the entire corpus.

To statistically validate the measurement, we checked how our final model predicts the human-coded labels of the validation set that was set aside. The Pearson correlation coefficient between the human-coded labels and the model predictions is 0.81, and the root mean squared errors is 0.53 (see app. fig. A3). Compared to other prediction practices (Park 2021; Park and Montgomery 2024) that used similar measurement processes, our model achieved a relatively high level of prediction performance, suggesting that it effectively captured the aspects of the latent trait we intended to measure.

In addition, for substantive validation, we present three statements with the highest, median, and lowest analytical scores in table 1. We can see that the most analytical statement makes the budget justification referring to the source of expert information; in the statement with the median score, the witness says that she does not have enough knowledge at the moment but would like to provide accurate information sourced from other bureaucrats in the future; lastly, the least analytical statement emphasizes the speaker's lack of information about what was being asked. These examples substantially validate our measurement. We present additional examples of analytical and nonanalytical statements and statistical validation metrics of the analytical score in appendix section D.

Figure 4 shows the distribution of analytical information within and across agencies. It is clear that the amount of

9. We constructed two document-level matrices: a term-document frequency matrix and a doc2vec matrix. For each matrix, we fit four learners: support vector machine, Kernlab's support vector machine, LASSO, and Gradient Boosting Machine. These choices are explained in app. sec. D.

Table 1. Sample Statements with Varying Levels of the Analytical Score

Analytical Score	Speech	Witness
100	"Preventing the extinction of this unique frog will require the restoration of ponds and surrounding habitats and the reintroduction of frogs from the one remaining population. In addition, as part of the $3.7 million increase for Pacific Northwest salmonid, a $0.6 million program increase will help the Service meet its obligations as set forth by the Federal Columbia River Power System Biological Opinions. . . ."	Steven A. Williams, Director of Fish and Wildlife Service, US Dept. of the Interior
30.8	"So let me be very explicit about this. . . . I would love to get you that information. Again, I am not an enforcement agency. So I am not as familiar with these details as some of my other assistant secretaries might be. So I would love to keep the door open so that I could get you more accurate information on that from our attorneys and the people in the enforcement agencies."	Jane Oates, Assistant Secretary for the Employment and Training Administration, US Dept. of Labor
0	"I guess I don't know what to think of it. I was surprised by it. I believe that they are friends and—but I don't know."	Mary Schapiro, Chairman, US Securities and Exchange Commission

Note. The first statement was made in a hearing held by the Committee on Resources in 2002 on Fish and Wildlife Service Agencies and their budget requests for 2003. Due to the space limit, only the front part of this statement is shown. The second statement is part of a Judiciary Committee hearing held in 2011 on H-2A visa program. The third statement is from a joint hearing between subcommittees of the Financial Services Committee and the Oversight and Government Reform Committee held in 2011 on how conflicts of interest are handled within US Securities and Exchange Commission.

within-agency variation still exceeds the amount of across-agency variation for both the executive departments and other agencies.[10]

Legislator–witness interactions

To test the effect of partisan incentives on witnesses' information sharing with members of Congress, we aggregate the data obtained from hearing transcripts by legislator–witness pairs. In this pair-level dataset, the unit of observation is a pair of a legislator and a witness who interacted in a House committee hearing, and it covers the period from 1997 to 2014. We define that an interaction occurred if a member statement is immediately followed by a witness statement.[11] We compute the average analytical score measured based on a witness's testimonies for each pair.

In total, we have 61,802 pairs of bureaucratic witness–member observations and 87,216 pairs of nonbureaucratic witness–member observations from 6,686 hearings. For bureaucratic witnesses, on average, we have nine pairs of witness–member interactions per hearing, ranging from 1 to 102. On average, in a given hearing, two bureaucrats were invited as witnesses and seven members asked them questions. For nonbureaucratic witnesses, on average, we have 11 pairs of witness–member interactions per hearing, ranging from 1 to 131. On average, four nonbureaucratic witnesses were invited and five members asked them questions.

THE EFFECT OF PARTISAN ALIGNMENT ON INFORMATION SHARING

In this section, we empirically examine how incentives underlying the partisan alignment between Congress and the executive branch, as well as the type of bureaucrat, drive the amount of analytical information that bureaucrats share. Using the pair-level data, we estimate the following regression:

$$\begin{aligned}\text{Analytical Information}_{bmh} = {} & \beta_1 \text{President Out} - \text{Partisan}_m \\ & + \beta_2 \text{Member Question}_{bmh} + \Gamma X_{bmh} \\ & + \alpha_a + \alpha_c + \alpha_i + \alpha_p + \varepsilon_{bmh}, \end{aligned} \quad (3)$$

10. There are 55 non–executive department agencies in our sample. For illustrative purposes, in fig. 4*B*, we plot the distribution of analytical information scores for the 15 agencies with the highest number of bureaucrats testifying in hearings.

11. Since witnesses speak only when they are given a chance to speak, most witnesses' statements are preceded by a member's statement. This feature facilitates analyzing dyadic interactions between a member and a witness in a hearing.

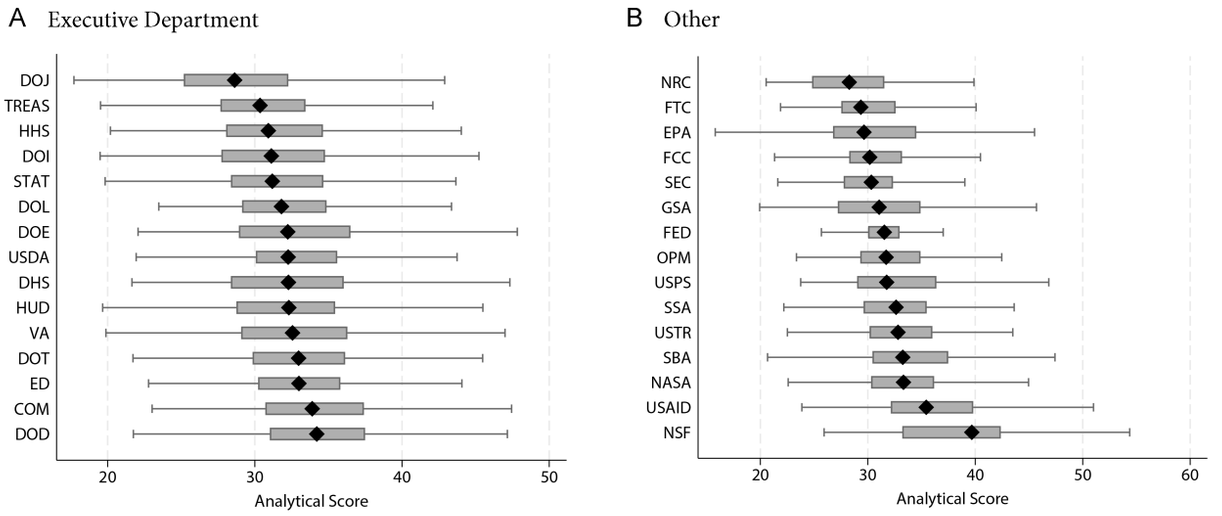


Figure 4. Analytical score distributions by agencies. The bars' ranges indicate the minimum and maximum values, the boxed area indicates the interquartile range, and the black squares indicate each agency's median analytical score.

where the subscripts indicate bureaucrat b, member m, hearing h, agency a, committee c, issue i, and president p. The unit of observation is bureaucrat–member pair conditional on a member asking at least one question of a bureaucrat in a given hearing. The outcome variable $Analytical\ Information_{bmh}$ is the average analytical score of bureaucrat b's testimony in response to member m's question in hearing h.[12] To capture partisan misalignment at the member–bureaucrat level, we use the *President Out-Partisan* variable, which equals 1 if the questioning member's party affiliation is different from the party of the president and equals 0 otherwise. We also include an interaction term between *President Out-Partisan* and *Political Appointee* variables to see whether the partisan dynamics are more salient when politically appointed bureaucrats interact with the president's out-partisan members.

Crucially, members' statements preceding a witness's testimony can shape the witness's response, and the manner in which legislators ask questions can be correlated with the variables that capture the partisan alignment. To account for this, we measure various features of members' questioning styles (*Member Question*). Due to the public nature of congressional hearings, sometimes legislators may intend to use a hearing to send political messages and gain media attention rather than to elicit analytical information from a witness. Thus, first, we address this underlying incentive of legislators by controlling for the grandstanding score, which was constructed to measure the intensity of political messages conveyed in each statement that members made in committee hearings (Park 2021).[13]

Legislators' grandstanding incentives, however, may not necessarily preclude the exercise of their duty to check facts; they may pursue both at the same time (e.g., Eldes et al. 2024). Therefore, to add another control for legislators' intent when they interact with a witness, we also measure members' incentive to seek information by counting the number of keywords that are indicative of analytical information in member statements. We use the analytical keywords introduced in Ban et al. (2023).

We also measure and control for the sentiment in members' questions using pysentimiento, which is a Python package that provides a pretrained BERT-based sentiment classifier (Pérez, Giudici, and Luque 2021). For each analyzed statement, pysentimiento generates three continuous scores for each statement: positive, neutral, and negative. Finally, we measure the number of words in each member's question because the length of questions may correlate with our measures of members' questioning tone. As these measures were constructed for each member statement, we compute the average values of these measures for each legislator–witness pair in a hearing. Table B1 in the appendix shows the mean

12. We trimmed the observations with an outlier value of the outcome measures below 1% and above 99%. Including these outliers does not change the results.

13. The operational definition of grandstanding includes (1) denouncing or praising a person or an institution, (2) taking a position on a policy (which includes a subjective interpretation of a policy-relevant situation), or (3) asking questions designed to attack or embarrass a witness. The study defines nongrandstanding statements as those that (1) offer an objective description of a policy-relevant situation or (2) question a witness for the purpose of fact-finding or obtaining an expert opinion. The score was constructed using a crowd-sourced supervised learning method at the statement level and ranges from 0 to 100.

value for each variable that captures the questioning styles of members.

X_{bhm} includes various bureaucrat-level, member-level, and hearing-level control variables (e.g., the number of witnesses in a hearing). To capture the partisan characteristics of a bureaucrat, we include the *Political Appointee* variable, which is coded as 1 if a bureaucrat is a political appointee requiring Senate confirmation (PAS).[14] We also note whether a questioning member is a committee chair or a ranking member, the member's majority party status, and their legislative effectiveness score from Volden and Wiseman (2014). Also, we include the members' ideological extremism, measured by the absolute difference between a member's DW-NOMINATE score and the decade-level median DW-NOMINATE score for each party.[15]

Given the way that hearings are structured, the order in which bureaucrats answer members' questions can also affect the amount of analytical information bureaucratic witnesses share with the committee. For example, suppose the first question to a witness comes from a member of the president's party, and suppose the bureaucrat, in response to that question, gives all the analytical information she prepared to share. The next question might come from a presidential out-partisan member. There may be little analytical information left to share. This situation raises the possibility of a correlation between analytical information and how many questions have already been asked. To address this issue, we extract the information about the order of each member–witness dyad that appears in each hearing (*Dyad Order*) as well as the order of the first dyadic member–witness interaction for each witness in a given hearing (*Witness Response Order*). In the regression, we control for the order of when the witness response is provided.[16]

We include bureaucrats' agency fixed effects (α_a) to control for agency-specific characteristics. We also include issue fixed effects (α_i) based on major issue codes from the Policy Agenda Project (Baumgartner and Jones 2015) and committee fixed effects (α_c). To control for time trend, we include president fixed effects (α_p).[17] ε_{bmh} is clustered at the hearing level. Appendix table B2 presents the summary statistics on the variables included in the regression analysis.

Bureaucratic witnesses and provision of analytical information

First, we examine how partisan alignment affects the information sharing between bureaucratic witnesses and members of Congress at the pair level. Table 2 presents the results of estimating equation 3 for legislative hearings and oversight hearings. Columns 1 and 4 show the results without including members' questioning styles.[18] Across both types of hearings, partisan misalignment between members of Congress and bureaucrats affects the amount of analytical information that bureaucrats share during hearings. Overall, bureaucrats provide less analytical information when they interact with legislators who are president out-partisan.

This pattern is more pronounced when political appointees are responding to a member who is an out-partisan of the president in oversight hearings. Compared to legislative hearings, bureaucratic agencies may be on political defense more so in oversight hearings, when Congress tends to examine the performance of a bureaucratic agency and discuss the potential problems or corrections that the agency needs to make in its implementation of government programs and policies. It is unsurprising, then, that the behavior of political appointees with out-partisans is more robust in oversight hearings.

As discussed previously, the negative association between partisan misalignment and the provision of analytical information by bureaucrats could be driven by legislators' questioning styles. As table B1 in the appendix shows, under divided government, president out-partisan members ask less analytical questions and their tone is more negative. These patterns of questioning styles also could influence how much analytical information bureaucratic witnesses share during their testimonies. Columns 2 and 5 show the results

14. Among 3,409 political appointees who appeared in hearings in our dataset, 73% are PAS, 24% are political appointees without Senate confirmation required (PA), and 3% are Schedule C (excepted service nonpermanent) appointees. For robustness, we also run the regression with defining the *Political Appointee* including all types of political appointees, and the results are presented in app. table B3. These results are similar to the main results.

15. Using the absolute distance from zero to measure a legislator's ideological extreme is inappropriate because the scale 0 is a latent variable and it has no substantive meaning. Instead, we use a global average over multiple years for each party and treat a further deviation from the global average as more extreme. We use a decade (10 years) to calculate a global mean and calculate the absolute distance between a legislator's DW-NOMINATE score and the median of the legislator's party at the decade level to measure the legislator's ideological extremism.

16. App. figs. A4 and A5 show the average analytical information by the variables *Dyad Order* and *Witness Response Order*.

17. In some regressions, we use the *Divided Government* variable that varies at the Congress level; to estimate the effect of this variable while controlling for time trend, we use president fixed effects. Using the Congress fixed effect when we do not need to estimate the *Divided Government* variable produces similar results.

18. We understand that variables of members' questioning styles are post-treatment, but we include these in subsequent regressions in order to control for members' behavior.

Table 2. Bureaucratic Witnesses' Provision of Analytical Information

	Legislative			Oversight		
	(1)	(2)	(3)	(4)	(5)	(6)
President out-partisan	−1.404***	−1.221***	−1.188***	−1.556***	−1.299***	−1.162***
	(.142)	(.132)	(.155)	(.162)	(.150)	(.167)
Political appointee	.0382	−.437*		.477*	−.124	
	(.211)	(.199)		(.218)	(.198)	
President out-partisan × Political appointee	−.407	−.163	−.0851	−.886***	−.451*	−.628*
	(.233)	(.215)	(.255)	(.240)	(.222)	(.247)
Majority party	−.189	−.198	−.109	−.180	−.0363	−.0481
	(.132)	(.122)	(.144)	(.144)	(.134)	(.146)
Committee chair	3.504***	3.263***	2.487***	3.825***	3.753***	2.615***
	(.186)	(.171)	(.209)	(.209)	(.191)	(.214)
Ranking member	.0874	.323	−.0987	1.053***	1.454***	.924***
	(.261)	(.243)	(.291)	(.256)	(.238)	(.271)
Ideological extreme	−1.862***	−.487	−.441	−2.576***	−1.282**	−.698
	(.436)	(.412)	(.473)	(.463)	(.426)	(.472)
LES	−.0776	−.0643	−.0884	−.0112	−.0220	−.0366
	(.0422)	(.0391)	(.0470)	(.0460)	(.0415)	(.0446)
Subcommittee	.385	.491**		1.367***	1.514***	
	(.200)	(.184)		(.221)	(.200)	
Referral hearing	−.402	−.567**		−.337	−.497	
	(.222)	(.208)		(.362)	(.334)	
Number of witnesses	.126***	.112***		.115***	.128***	
	(.0218)	(.0202)		(.0224)	(.0227)	
Dyad order	−.171***	−.137***	−.239***	−.0779***	−.0641***	−.185***
	(.0115)	(.0103)	(.0162)	(.0101)	(.00904)	(.0129)
Grandstanding score		−.126***	−.102***		−.105***	−.0832***
		(.00798)	(.00959)		(.00825)	(.00926)
Share of positive tone		6.838***	5.686***		6.353***	5.106***
		(.265)	(.319)		(.287)	(.325)
Share of negative tone		−1.574***	−.653		−1.259**	−.939*
		(.372)	(.445)		(.384)	(.426)
Analytical word count		.167***	.164***		.154***	.136***
		(.0175)	(.0218)		(.0176)	(.0204)
(ln) Number of legislator words		1.794***	1.767***		1.772***	1.902***
		(.150)	(.184)		(.167)	(.189)
Agency, committee, issue, president FEs	Yes	Yes		Yes	Yes	
Witness-hearing FE			Yes			Yes
Mean outcome measure	33.3	33.3	33.3	33.5	33.5	33.5
n	30,117	30,117	30,117	27,401	27,401	27,401
Adjusted R^2	.098	.190	.325	.112	.202	.333

Note. Standard errors are clustered at the hearing level. FE, fixed effect.

* $p < .05$.

** $p < .01$.

*** $p < .001$.

when we include the variables that capture members' questioning styles. The results show that members' questioning styles are indeed correlated with the amount of analytical information in bureaucrats' testimonies. When members ask questions with higher grandstanding scores or a more negative tone, bureaucrats tend to provide less analytical information. In contrast, when members ask questions that have a more positive tone, more analytical words, and more words in total, bureaucrats' testimonies tend to include more analytical information.[19] The member–witness order (*Dyad Order*) has a consistently negative sign, suggesting that less analytical information is shared by bureaucratic witnesses as the hearing progresses.[20]

It is interesting to note that even after controlling for members' questioning styles, the coefficient on the variable *President Out-Partisan* is negative and statistically significant, although the magnitude of the coefficients becomes slightly smaller. Of course, the set of control variables for the members' questioning styles may not fully capture members' intentions. However, the result could suggest that bureaucrats may control the amount of analytical information they share with committee members depending on whether they interact with president co-partisan or president out-partisan legislators. The results in table 2 also show that bureaucrats' testimonies contain significantly more analytical information when they are questioned by the committee chair and ranking member. In addition, bureaucrats' statements contain more analytical information in hearings held at the subcommittee level as opposed to the full committee level, which are hearings that feature more witnesses.

Although we include agency, issue, and committee fixed effects to control for the selection issues, the frequency of bureaucratic invitations and the types of bureaucrats who are invited could be different under divided versus unified governments (Ban et al. 2023). In addition, the identity of the bureaucrats who receive questions from president co-partisan members versus president out-partisans could be different. To address this potential selection problem, we exploit our pair-level data, which allows us to estimate equation 3 with witness-hearing fixed effects. We examine how the same witness in a given hearing responds differently to different members.[21] By including witness-hearing fixed effects, we are unable to measure the effect of some variables that do not vary at the witness-hearing level, such as political appointee, but this exercise is the most rigorous estimation of how bureaucrats respond to partisan misalignment with politicians in hearings. We use the same specification as before but with witness-hearing fixed effects. Because we examine within-bureaucrat variation in a given hearing, we do not include other fixed effects. Columns 3 and 6 of table 2 show the results. We see consistent results from the specifications including witness-hearing fixed effects: The same bureaucrat's response contains a lower level of analytical information when she responds to members who are president out-partisans, and this effect is more salient when the bureaucrat is a political appointee appearing in oversight hearings.[22]

In terms of the substantive effect, each bureaucrat provides 4.6% less analytical information when questioned by members who are president out-partisans, and this reduction increases to 7.2% when political appointees interact with president out-partisans in hearings.[23] Given that 12,704 bureaucrats were invited to congressional hearings during the period of our study (1997–2014), and 27% of them were Senate-confirmed political appointees, at the aggregate level, this is a significant reduction in the amount of analytical information that bureaucrats share with members of Congress.

Additional analyses of bureaucratic witnesses

In this section, we provide additional analyses of bureaucratic witnesses and the interbranch flow of analytical information in hearings that may enhance our understanding of the dynamic interactions between legislators and bureaucrats. Our analyses in this section parse how differences across bureaucrats—such as the types of their agencies or how ideologically close their agencies are to the president—and differences across issues and committees—such as the issue polarization and types of committees—moderate the effect of partisan alignment on the analytical information shared.

First, we conduct separate analyses for bureaucrats who come from the executive departments and other agencies. Because executive departments are subject to more direct executive control by the president, who can hire and fire

19. The reference category is the share of neutral tone.

20. If we instead include the variable *Witness Response Order* to capture the effect of the response order within a witness, we also observe a negative coefficient, and including this variable does not change the main result. See app. table B4.

21. A unique witness ID is assigned at the witness-hearing level. If the same bureaucrat appeared in different hearings, the bureaucrat is assigned different witness IDs.

22. In app. table B5, we further confirm that the negative coefficient on the interaction term *President Out-Partisan* × *Political Appointee* is more salient under divided government when president out-partisan legislators are the majority party members in Congress.

23. This calculation is based on the results in column 4 of table 2. Given the mean outcome measure of analytical information of 33.5, the coefficients on the variables *President Out-Partisan* and *President Out-Partisan* × *Political Appointee* indicate the reduction of 3.1% (−1.556/33.5) and 5.5% ([−1.556–0.886]/33.5).

department heads and other political appointees at will, the partisan dynamics for political appointees may differ between executive departments and other agencies. Table B6 in the appendix shows that a statistically significant negative relationship for the interaction term *President Out-Partisan* × *Political Appointee* is observed only among executive department bureaucrats in oversight hearings.[24]

Second, our main analysis focuses on the partisan alignment between Congress and the president, and we assume that bureaucrats are under the control of the president. However, bureaucratic agencies exhibit various ideological leanings (Richardson et al. 2018), and individual bureaucrats have their own partisan leanings (Spenkuch et al. 2023). This implies that some executive agencies are more aligned with the president than others depending on the partisanship of the president. We use the measure for the agency ideology from Richardson et al. (2018) to capture the partisan alignment between the president and the agency. Specifically, we divide agencies into those aligned and misaligned with the president in terms of ideology and examine whether the partisan dynamics of analytical information provision are more salient for bureaucrats from presidentially aligned agencies.[25] Table B8 in the appendix shows that bureaucrats from agencies that are aligned with the president are less likely to provide analytical information to president out-partisan committee members.

Third, we examine whether the effect of partisan alignment on bureaucrats' provision of analytical information varies by issue polarization and committee type. A full discussion of these heterogeneous effects can be found in appendix section C. Overall, we find that bureaucrats provide significantly less analytical information when they appear in hearings that address highly polarized issues. Moreover, the negative effect of partisan misalignment is larger in prestigious and policy committees than in constituent service committees (Deering and Smith 1997).

Overall, the additional analyses in this section show that when the partisan nature of the legislator–bureaucrat alignment is more salient—such as when the bureaucrat is from an agency more closely aligned ideologically with the president or when the issue is highly polarized—the effect of facing an out-partisan legislator on the amount of analytical information shared is stronger.

Placebo test: Nonbureaucratic witnesses

Finally, we examine whether these partisan misalignment effects are unique to bureaucrats or are shared by other witnesses as well. Are nonbureaucratic witnesses, who are not employed in the executive agencies, affected by the interbranch relationship between the legislative and executive branches? As a "placebo" test, we perform the same analysis on all other nonbureaucratic witnesses.[26] In total, we have 87,261 pairs of interactions between committee members and nonbureaucratic witnesses for the period of 1997 to 2014.

We measure the analytical score for all other witnesses, create a pair-level dataset for nonbureaucratic witnesses, and run the same specification as in equation 3. Table 3 presents the regression results for nonbureaucratic witnesses.[27] There are no statistically significant effects of the variable *President Out-Partisan* on the sharing of analytical information by nonbureaucratic witnesses. This is in stark contrast to the previous results for bureaucratic witnesses, where the same variable has statistically significant, negative effects. Overall, this placebo test suggests that the interbranch relationship has a unique effect on the behavior of bureaucratic witnesses in sharing analytical information with committee members.

Partisan dynamics and collective information acquisition

In the previous sections, we show that bureaucratic witnesses provide less analytical information in their testimony when they respond to questions from members of the president's own party. We also show that we do not find a similar partisan dynamic when it comes to nonbureaucratic witnesses. These results speak to information transmission at the pair level between two individuals—a legislator and a witness. However, given the public nature of hearings and the fact that members can observe how their colleagues, from both parties, interact with all witnesses in a hearing, this raises the question of how hearings serve members of Congress in the collective acquisition of information and how partisan dynamics affect this process. Even if bureaucratic witnesses provide less analytical information to members who are presidential out-partisans, if these legislators are more actively engaged in hearings by

24. App. table B7 presents the result when we classify the EPA as part of the executive department given the president's control over the agency. The main results do not change.

25. Based on the measure of agency ideology developed by Richardson et al. (2018), we label agencies as Democratic-leaning if the ideology is below the median (−0.1256831). These agencies are coded as aligned with the Democratic president. For Republican presidents, we coded these agencies as aligned with the president if the agency's ideology is above the median ideology.

26. We exclude congressional staff, members of Congress, or affiliates of congressional organizations to focus on nongovernmental witnesses.

27. With nonbureaucratic witnesses, we include witness type fixed effects (16 different types, such as corporations, nonprofits, and think tanks and academics) instead of agency fixed effects. A full regression result is provided in app. table B9.

Table 3. Nonbureaucratic Witnesses' Provision of Analytical Information

	Legislative			Oversight		
	(1)	(2)	(3)	(4)	(5)	(6)
President out-partisan	−.169	−.0362	.0582	−.176	−.158	−.100
	(.106)	(.146)	(.218)	(.159)	(.216)	(.310)
Witness type, issue, committee, president FEs	Yes			Yes		
Witness-hearing FE		Yes	Yes		Yes	Yes
Bureaucrat appeared in hearing			Yes			Yes
Mean outcome measure	35.3	35.3	35.9	35.5	35.5	35.9
n	52,217	52,217	27,383	28,168	28,168	14,241
Adjusted R^2	.222	.350	.352	.232	.359	.358

Note. Standard errors are clustered at the hearing level. Columns 3 and 6 present the results for the hearings where nonbureaucratic witnesses appeared along with at least one bureaucratic witness. FE, fixed effect.

* $p < .05$.

** $p < .01$.

*** $p < .001$.

asking more questions of nonbureaucratic witnesses under divided government, this may generate more analytical information for Congress at the aggregate level.

To address this question, we created a hearing-level dataset that included the total number of question–answer pairs between a member and a witness (i.e., which captures how actively members engage with witnesses) and the total amount of analytical information weighted by the number of words from witness testimony. We also created these variables separately for bureaucratic and nonbureaucratic witnesses.

We use the presence of divided government as a measure of partisan alignment between the legislative and executive branches at the aggregate hearing level. Under divided government, when the two branches are represented by opposing parties with different electoral and partisan incentives, the flow of policy-relevant information may be hampered by strategic interactions between legislators and bureaucrats. Divided government presents a more difficult policymaking environment, as politicians between the legislature and the executive have opposing leverage and face institutional impediments in the branch their party does not control. This challenging institutional environment has been the basis for theoretical arguments explaining the complications that divided government brings to policymaking (Coleman 1999; Howell et al. 2000).

Using our hearing-level data, we run a regression to examine how divided government correlates with collective information acquisition. Table 4 presents the results.[28] First, under divided government, members of Congress—particularly presidential out-partisans—ask more questions of both bureaucratic and nonbureaucratic witnesses compared to presidential co-partisans. Second, we find that divided government is not correlated with total analytical information at the hearing level. While bureaucratic witnesses provide less analytical information under divided government, this decline is offset by increased analytical contributions from nonbureaucratic witnesses. The key driver of this pattern is out-partisan members' greater questioning of nonbureaucratic witnesses, while co-partisan members ask significantly fewer questions—perhaps due to being in the House minority—reducing analytical input from both groups. These opposing effects cancel out, leaving the aggregate level of analytical information in hearings unchanged.

Overall, this set of results highlights how partisan dynamics differentially affect Congress's ability to acquire analytical information at the individual and collective levels. Although the partisan misalignment reduces the transmission of analytical information between bureaucratic witnesses and out-partisan members at the pair level, the presence of nonbureaucratic witnesses and increased questioning by out-partisan members during divided government leads to no discernible effect on the transmission of analytical information

28. A detailed analysis by witness type is in app. tables B10 and B11.

Table 4. Divided Government and Information Transmission at the Aggregated Level

	No. Question–Answer Pair			Analytical Information		
	(1) Total	(2) Co-Partisan	(3) Out-Partisan	(4) Total	(5) Bureaucrat	(6) Nonbureaucrat
Divided government	2.813***	−2.974***	5.787***	−4.419	−51.82**	47.40
	(.365)	(.344)	(.424)	(27.44)	(17.90)	(23.49)
Issue, committee, president FEs	Yes	Yes	Yes	Yes	Yes	Yes
Mean outcome measure	16.8	7.7	9.1	1,450.3	492.0	958.3
n	10,753	10,753	10,753	10,753	10,753	10,753
Adjusted R^2	.423	.354	.449	.461	.220	.452

Note. The unit of observation is hearing. Standard errors are clustered at the committee level. Control variables (oversight hearing, subcommittee, referral hearing, number of witness) are included but not presented here. FE, fixed effect.

* $p < .05$.

** $p < .01$.

*** $p < .001$.

at the aggregate level. Our findings provide a nuanced understanding of the effect of divided government on information transmission in hearings.

CONCLUSION

In democracies, the people who make the laws are not those who implement the laws. This division of labor results in disparities in the levels of knowledge and expertise between legislators and bureaucrats. Information about policy implementation and its costs and consequences is gained through on-the-ground experience in the bureaucracy; legislators must rely on searching for and acquiring that information. Compared to members of Congress, bureaucrats are closer to policy implementation and thus gain more expertise and specialized information. Scholars have documented and theorized about the informational advantage held by bureaucrats, but the question of what affects the exchange of information between bureaucrats and Congress has remained unanswered, especially empirically.

Using a new dataset that provides—for the first time—the federal agency affiliation, appointment type, and agency-level characteristics for each bureaucrat who testified in Congress over the span of decades of congressional history, we find that partisan dynamics exert a nuanced influence on the transmission of analytical information during congressional hearings. At the pair level between an individual bureaucrat and legislator, bureaucratic witnesses offer less analytical information when questioned by members of the president's party, a pattern not observed with nonbureaucratic witnesses. At the hearing level, divided government prompts increased engagement from members of the president's opposing party, who ask more questions, thereby compensating for the reduced input from bureaucratic witnesses. Consequently, the overall level of analytical information transmitted in hearings remains stable despite these partisan influences. This understanding reveals that while individual interactions between members and witnesses are affected by partisanship, the collective acquisition of information in Congress is maintained through the dynamics of questioning under divided government.

Our research and new data provide a refreshed foundation for continued work on the interbranch relationship between bureaucrats and Congress. As we show, the partisan alignment between Congress and bureaucrats has important implications for the information that members of Congress collect through hearings. Congress relies heavily on information provided by the executive branch to make policy decisions with far-reaching consequences—especially concerning complex scientific issues facing society today, such as climate change or pandemic response. Thus, understanding what affects information sharing within government—between bureaucrats, one of Congress's best sources of policy information, and Congress—is paramount.

This study lays the groundwork for future research to further unpack bureaucrats' individual incentives. Our findings suggest that partisan misalignment leads bureaucrats to

provide less analytical information to legislators, potentially reflecting strategic considerations. Future work could explore whether this behavior is driven by policy preferences, career incentives, or personal ideology. Additionally, research should examine how partisan misalignment in hearings influences legislators' broader strategies for acquiring executive branch information. Congress can request various reports, from agency activity summaries to studies on emerging issues like artificial intelligence (Egar 2020). A promising extension of this study would explore the legislative tools Congress employs to access executive information and how partisan alignment between the branches shapes these choices.

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